

WEEK 1 · FINAL EXAM · ALL NINE COURSES

Week 1 Final Exam

Name: _____

Date: _____

How to play: Read each question twice, read every choice, cross out the wrong ones, then put an X in the box next to your answer. Get **17 of 24** right to pass and earn your Week 1 diploma.

1. Which note best follows the Freshman Seminar rule for note-taking?

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- A. Lesson 1 was about the isolette; it took about 20 minutes and there was a quiz at the end.
- B. An isolette, also called an incubator, looks like a see-through spaceship bed with a clear hood.
- C. Isolette, incubator, skin probe, servo control, humidity, kangaroo care, hypothermia, brown fat.
- D. Isolette — the warm box; its skin probe adjusts the heat automatically.

2. What does Elizabeth Blackwell's story teach a student?

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- A. Medical schools in the 1840s welcomed women as long as they had good grades.
- B. Being a good doctor mostly requires being popular with your classmates.
- C. Do the work and let your record speak, even when you're not welcome.
- D. The surest way into medical school is to know someone important there.

3. Glucose (sugar) is $C_6H_{12}O_6$. How many atoms are in one molecule?

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- A. 12
- B. 6
- C. 3
- D. 24

4. Which gas makes up most of the air you breathe?

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- A. Water vapor
- B. Carbon dioxide
- C. Nitrogen
- D. Oxygen

5. Which part of a cell burns sugar with oxygen to make energy?

- ☐ A. Nucleus
- ☐ B. Membrane
- ☐ C. Mitochondria
- ☐ D. Cytoplasm

6. One cell goes through 5 rounds of division (doubling each time). How many cells now?

- ☐ A. 32
- ☐ B. 10
- ☐ C. 25
- ☐ D. 16

7. Pushing with the same force through a needle's point instead of your fingertip...

- ☐ A. makes the pressure higher, because metal weighs more than skin does.
- ☐ B. makes the pressure much lower, because the area is much smaller.
- ☐ C. leaves the pressure the same, because the push is exactly the same.
- ☐ D. makes the pressure much higher, because the area is much smaller.

8. 1 mmHg is about 1.36 cmH₂O. Why is the mercury unit the bigger push?

- ☐ A. The mercury column is always read at a warmer temperature than water.
- ☐ B. Mercury is about 13.6 times heavier than water, so lifting it takes more push.
- ☐ C. Centimeters are about 1.36 times longer than millimeters on a ruler.
- ☐ D. Water is about 13.6 times heavier than mercury, so it needs less push.

9. A baby weighed 1,100 g on day 1 and 1,240 g on day 8. What is the rate of change?

- ☐ A. 2 grams per day
- ☐ B. 20 grams per day
- ☐ C. 140 grams per day
- ☐ D. 14 grams per day

10. A weight graph shows a flat, level line for a week. That means...

- ☐ A. the baby slept through most of the week.
- ☐ B. the scale stopped working during the week.
- ☐ C. the baby grew at a steady, healthy pace.
- ☐ D. the baby's weight did not change all week.

11. What is the mean of 120, 130, 140, and 150?

- ☐ A. 135
- ☐ B. 540
- ☐ C. 140
- ☐ D. 30

12. A baby's Apgar score is 4 at one minute. What does that mean?

- ☐ A. The baby is doing well and needs only routine newborn care today.
- ☐ B. The baby will have lifelong problems, and the family should be told so.
- ☐ C. The baby may need some help right now – rubbing, warming, perhaps some oxygen.
- ☐ D. The score must be a mistake, because healthy babies always score 9 or 10.

13. How many chromosomes does a baby receive from each parent?

- ☐ A. 46
- ☐ B. 23
- ☐ C. 92
- ☐ D. 2

14. A boy and a girl born together are...

- ☐ A. identical twins – they came from two eggs that merged into one.
- ☐ B. fraternal twins – they came from two separate eggs.
- ☐ C. not twins at all, because twins are always the same sex.
- ☐ D. identical twins – they came from one egg that split in two.

15. In the “Cat in the Hat” study, what did the newborns DO that showed they remembered the story?

- ☐ A. They fell asleep only when the familiar story was being read to them.
- ☐ B. They said the first words of the story by the time they were three months old.
- ☐ C. They cried whenever the unfamiliar story was read to them instead.
- ☐ D. They sucked on a pacifier in the pattern that played the familiar story.

16. Why do modern NICUs dim the lights and lower the noise?

- ☐ A. A preemie's brain is wiring itself, and calm, sleep, and gentle input help that wiring.
- ☐ B. Bright light slowly damages the clear plastic hoods of the isolettes.
- ☐ C. Loud noise would wake the parents who sleep in the unit overnight.
- ☐ D. Nurses can read the glowing monitor screens more clearly in the dark.

17. Who spends the most hours at a NICU baby's bedside?

- ☐ A. The pharmacist
- ☐ B. The neonatologist
- ☐ C. The NICU nurse
- ☐ D. The social worker

18. "Regionalization" of newborn care means...

- ☐ A. naming each NICU after the region of the country it serves.
- ☐ B. keeping every baby at the hospital where it happened to be born.
- ☐ C. making all the hospitals in a region share one team of doctors.
- ☐ D. getting the sickest babies to the hospitals equipped to care for them.

19. Hemoglobin came up in Chemistry, Biology, and Genetics. Which statement is TRUE?

- ☐ A. It's a gene made of oxygen atoms, stored inside the nucleus of every red blood cell.
- ☐ B. It's a protein built from a gene, packed into red blood cells, and it grabs oxygen molecules.
- ☐ C. It's a chromosome that carries carbon dioxide from the cells back to the lungs.
- ☐ D. It's a molecule of oxygen that turns bright red when it enters a body cell.

20. Which pairing correctly matches each course with its big question?

- ☐ A. Calculus: what molecules are in a breath? Statistics: how do cells divide and double?
- ☐ B. Calculus: what's typical across many babies? Statistics: how fast is this one baby growing?
- ☐ C. Calculus: how fast is this one baby growing? Statistics: what's typical across many babies?
- ☐ D. Calculus: how is the NICU team organized? Statistics: how does a baby's brain wire itself?

21. Flashback: why are preemies on extra oxygen kept around 90-95% saturation instead of 100%?

- ☐ A. The pulse oximeter cannot read any number above 95 percent accurately.
- ☐ B. Oxygen above 95 percent is too expensive for most hospitals to provide.
- ☐ C. Babies begin breathing more slowly whenever their number reaches 100.
- ☐ D. Too much oxygen harms developing eyes and lungs, while too little is dangerous too.

22. Flashback: what does surfactant do?

- ☐ A. It carries oxygen molecules around in the blood.
- ☐ B. It breaks down bilirubin in the skin under blue light.
- ☐ C. It warms the air inside the isolette automatically.
- ☐ D. It coats the lungs' air sacs so they don't stick shut.

23. Flashback: why do so many newborns turn yellow?

- ☐ A. Their skin cells make extra yellow pigment during the first week of life.
- ☐ B. Milk contains a yellow dye that shows through a newborn's thin skin.
- ☐ C. The isolette's warming lights tint the skin during the first few days.
- ☐ D. Bilirubin from recycled red blood cells builds up faster than the new liver can clear it.

24. Flashback: what does the fontanelle (soft spot) let doctors do?

- ☐ A. Ultrasound the brain through a bone-free window in the skull.
- ☐ B. Measure the baby's blood pressure without needing a cuff or a line.
- ☐ C. Feed the baby milk without needing a feeding tube or a bottle.
- ☐ D. Listen to the heart and lungs without needing a stethoscope.

Score: _____ / 24

Badge earned? YES / NOT YET (need 17)